What is now implemented and tested

Five-fold outer participant CV + three-fold inner CV
Three matched preprocessing arms; Ridge trained on CN source labels only

Implemented
- Raw connectome input
- Fold-specific edge QC and scaling
- Training-only ComBat and alpha selection
- Inner-OOF age-bias calibration
- Saved folds, models and provenance

Scope
- DWI Ridge baseline
- Known supported acquisition batches
- Canonical clinical IDs and labels

Clinical source validity and acquisition protocols still require review.

Implemented pipeline: training boundaries are explicit

Raw 84 × 84 DWI → fixed per-scan symmetrization / upper triangle → participant outer split

OUTER TRAINING
Source-labeled CN rows only
↓ Inner grouped CV
Fit edge mask → ComBat → scaler → Ridge
Choose alpha from inner OOF MAE
Fit BAG ~ age using inner OOF CN BAG
↓ Refit selected pipeline on outer-training CN

OUTER HELD-OUT PARTICIPANTS
Use frozen edge mask, offsets and scaler
↓ Ridge prediction
Score raw predicted age against actual age
↓ BAG = prediction − age
↓ cBAG = BAG − training-estimated age trend
No held-out age enters feature transformation.

All three arms use identical folds and training-derived edge masks. Non-CN rows are evaluated only when their participants are held out.

How can age-informed ComBat avoid held-out ages?

Training model: $x = \text{intercept} + \beta \times \text{training age} + \text{batch offset} + \text{residual}$
Mean-only ComBat adjusts batch location; its batch scale is fixed at 1.

Additive held-out transformation: $x_{harmonized} = x - \sigma_{training} \times \gamma_{batch}$
The subject-specific age terms cancel; no held-out age is needed.

Three prespecified comparisons
1. No ComBat 2. Batch-only location/scale ComBat 3. Training-age mean-only ComBat
Sex, diagnosis, APOE and metabolic variables are never brain-age predictor features.

This is an additive alternative, not full age-conditioned location/scale correction. Biology preservation must still be assessed.

Validation checks: what was actually tested

1. Training/apply equivalence
Frozen transforms reproduce the ComBat training output, including age-trained mean-only mode.

2. Held-out information isolation
Changing outer-test ages leaves inner alpha choice and calibration unchanged.

3. Independent-sample transformation
A sample transforms identically alone or with altered companion test samples; fitted state stays unchanged. Unknown batches raise an error.

4. End-to-end provenance
Participant groups remain disjoint; each eligible scan gets one OOF prediction per arm; calibration uses only outer-training rows.

Analysis population and exclusions

Cohort	Eligible scans	Participants	CN scans
ADDECODE	86	86	79
ADNI	187	106	141
ADRC	159	159	109
HABS	487	319	394

Archived input universe: 1,234 scans. Retained: 919 scans, including 723 CN scans.
Excluded: 127 ADNI scans from batches with insufficient CN participant support; 188 ADRC scans with ambiguous clinical keys.
These restrictions define the current estimand; they are not arbitrary first-row resolutions.

Matched outer-OOF prediction results: CN reference samples

Preprocessing	MAE (years)	RMSE (years)	R^2	Pearson r
Batch-only ComBat	8.246	10.922	0.205	0.460
Training-age mean-only	6.112	7.875	0.587	0.766
No ComBat	6.017	7.735	0.601	0.775

Same 723 CN scans / 546 CN participants; identical outer and inner folds.
Raw age predictions are scored before BAG correction. These are descriptive comparisons, not a significance test or a basis for post-hoc outer-test model selection.
Changed eligibility and labels prevent direct comparison with the old benchmark.

Scanner reduction and biological retention: limited probes

Preprocessing	Mean scanner balanced accuracy	Mean sex balanced accuracy
Batch-only ComBat	0.053	0.805
Training-age mean-only	0.421	0.803
No ComBat	0.653	0.812

Fixed-regularization probes are fitted on outer-training CN and evaluated on held-out participants.
Site predictability is confounded by cohort/age differences; lower accuracy does not prove that only technical effects were removed.
Age prediction and residual cBAG–age dependence are reported separately.

Remaining work before a project-wide scientific claim

Resolve clinical and acquisition provenance
Verify HABS source diagnoses, ADRC visit matching, canonical identities, acquisition dates and ADRC protocol consistency.

Extend the validated boundaries
Apply the same folds and training-only preprocessing to GINE, fMRI and fusion; nest graph sparsity, architecture choice and early stopping.

Clinical validation
Complete T2D/HbA1c availability checks; generate modality-specific held-out cBAG; prespecify clinical tests.

Interpret results cautiously
Known-site restricted population, one split seed, provisional source labels. No claim of unseen-site generalization or full biological preservation.